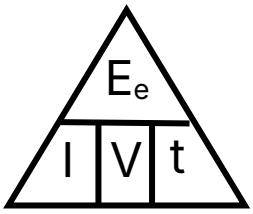
Calculating Electrical Energy, E_e

The formula for calculating electrical energy is:



E_e = IVt

Electrical Energy = Current x Voltage x time

- Current (I) is measured in Amps (A)
- Voltage (V) is measured in Volts
- Time (t) is measured in seconds (s)

Example

An electric hob in home economics has an operating voltage of 230 volts with a current of 5 amps. Calculate how much electrical energy has been used if the hob takes 5 minutes to heat up a pot of soup.

Remember to always convert your time into seconds.
Therefore 5 minutes x 60 = 300 seconds.

What are my knowns?
I = 5A V = 230V
t = 300 seconds

E_e = IVt

= 5 x 230 x 300

= 345,000 Joules

= 345 kJ

Questions

(*Answer to two significant figures*)

1. Find out how much electrical energy is transferred from a 230 volt power supply to a 12 volt battery which draws 5 amps for 2 minutes.

   E_e = IVt

   = 5 × 230 x (2 × 60)

   = 138,000J

   =**140,000J** to two significant figures (also 140kJ)

2. A hot air hand drier activates for 45 seconds once the switch is pressed. The drier operates from a 230V supply and draws a current of 10A. Calculate the amount of electrical energy used when the drier is operating.

   E_e = IVt

   = 10 × 230 × 45

   = 103,500J

   =**100,000** to two significant figures (also 100kJ)

3. A turbine is used to drive an 110V electrical generator. If the output current is 3A, how much electrical energy does the generator produce in 1 hour?

   E_e = IVt

   = 3 × 110 x (60 × 60)

   = 1,188,000J

   =**1,200,000** to two significant figures (also 1.2MJ)